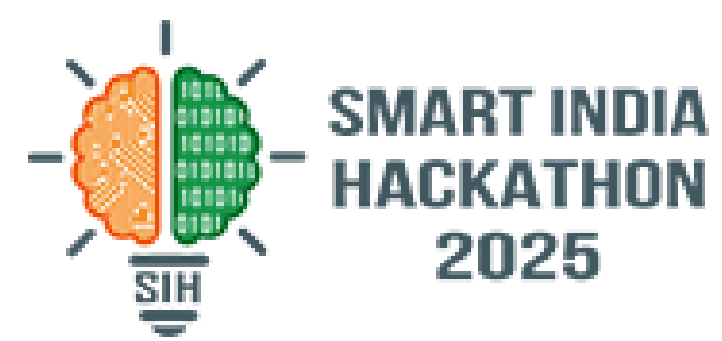
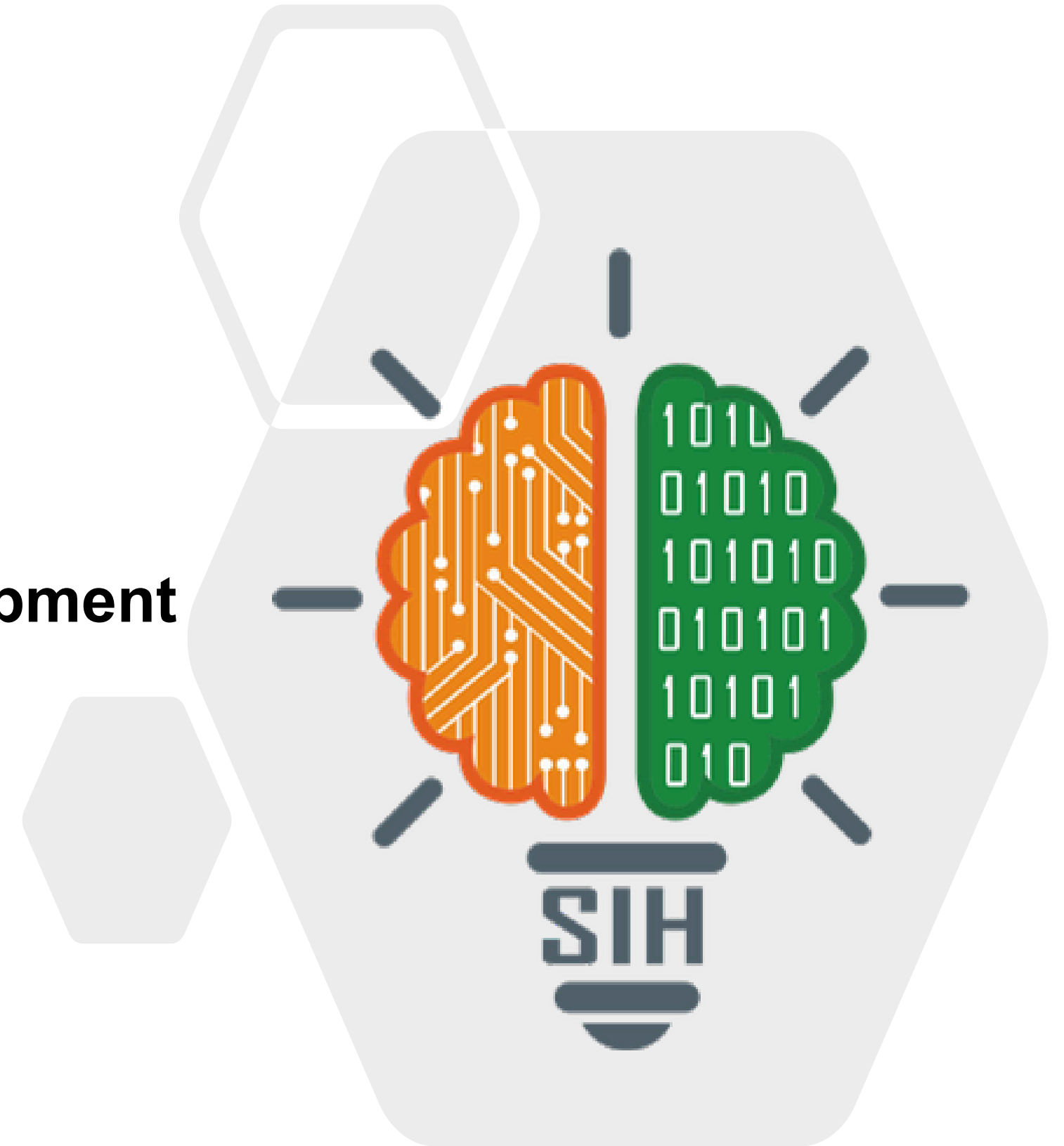




SMART INDIA HACKATHON 2025

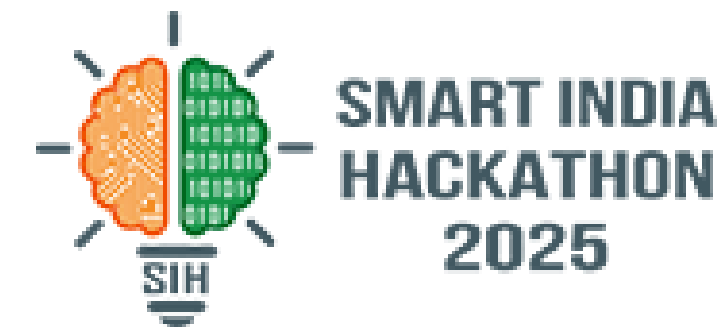


- **Problem Statement ID – SIH25126**
- **Problem Statement Title - Student Innovation**
- **Theme - Agriculture, FoodTech & Rural Development**
- **PS Category- Software**
- **Team ID - 89340**
- **Team Name - SteadyStride**





MittiMitra : Krishi Ka Sahayak



IDEA / SOLUTION

Build a comprehensive, AI-powered digital ecosystem that provides end-to-end support for farmers, from real-time monitoring and predictive analytics to automated financial services, ensuring accessibility for all through a multilingual, omnichannel interface

- **Dynamic Weather Forecasting & Alert System** : Use tomorrow.io for hyper-local weather data and Twilio for real-time weather alerts via SMS/voice, with an ML model(LSTM) for predictive fallback.[1]
- **Real-Time Mandi Price Intelligence** : Fetch live agricultural commodity prices from the AgMarknet API and Chart.js to create an intuitive, real-time dashboard visualizing crop price.[1]
- **Satellite-Driven Farm Health Monitoring** : Use Planet.com imagery and Leaflet.js to map NDVI crop health onto verified land plots.[1]
- **AI-Powered Crop Planning & Advisory** : An AI engine using real-time data to give tailored advice on crop rotation and planting.[2]
- **Blockchain-Enabled Agri-Finance & Automated Insurance**: Automated insurance uses data triggers. Blockchain verifies claims, and an AI model approves microfinance after crop failure.
- **Omnichannel, Multilingual Farmer Interface** : Build a multilingual chatbot. Use Twilio to provide SMS/voice access for button-phone users, ensuring complete digital inclusion.

PROBLEM RESOLUTION

- **Market & Weather Volatility**: Reduce financial risk with real-time weather alerts, predictive market price intelligence, and data-driven insights for informed decision-making.
- **Suboptimal Crop Management & Yield** :Our platform uses satellite imagery and AI to monitor crop health, providing precise advisories on rotation and inputs to maximize yield.
- **The Digital & Language Divide**: Our system is accessible to all via a multilingual bot and offers full functionality through SMS/voice for button-phone users.

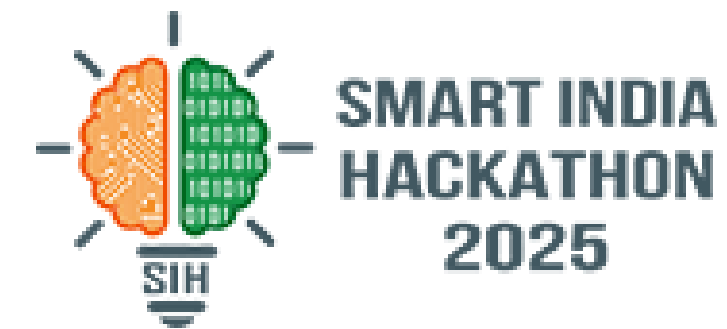
UNIQUE VALUE PROPOSITION

- End-to-End Farming Ecosystem
- Automated Financial Safety Net
- Radical Digital Inclusion
- Hyper-Personalized AI Advisory
- Real-Time Field & Market Intelligence





TECHNICAL APPROACH



Software Development:

• Frontend and User Interaction Layer:

1.Web Dashboard- Build using React+Next and visualization using Chart.js to display mandi price trends and Leaflet.js to render satellite NVDI data on farm plots.

2.Omnichannel Access: Twilio integration for Digital inclusion acts as a gateway for all SMS and voice based interactions.

• Authentication and Identity Management

1.Firebase Auth :Will manage all user sign-ups, logins, and profile security.

• Core Backend and Data Persistence

1.Microservices API: A set of backend services will serve as the core logic hub.

2.Primary Database:MongoDB will be the main database.Digital Ocean for Cloud deployment.

• Generative AI and Intelligence Layer

1.LLM for Advisory & Chatbot: A powerful LLM, likely based on a Transformer architecture(Llama)

2.Predictive Models: Specialized LSTM model trained on historical weather data to act as a predictive fallback.

• Blockchain & Financial Services Layer

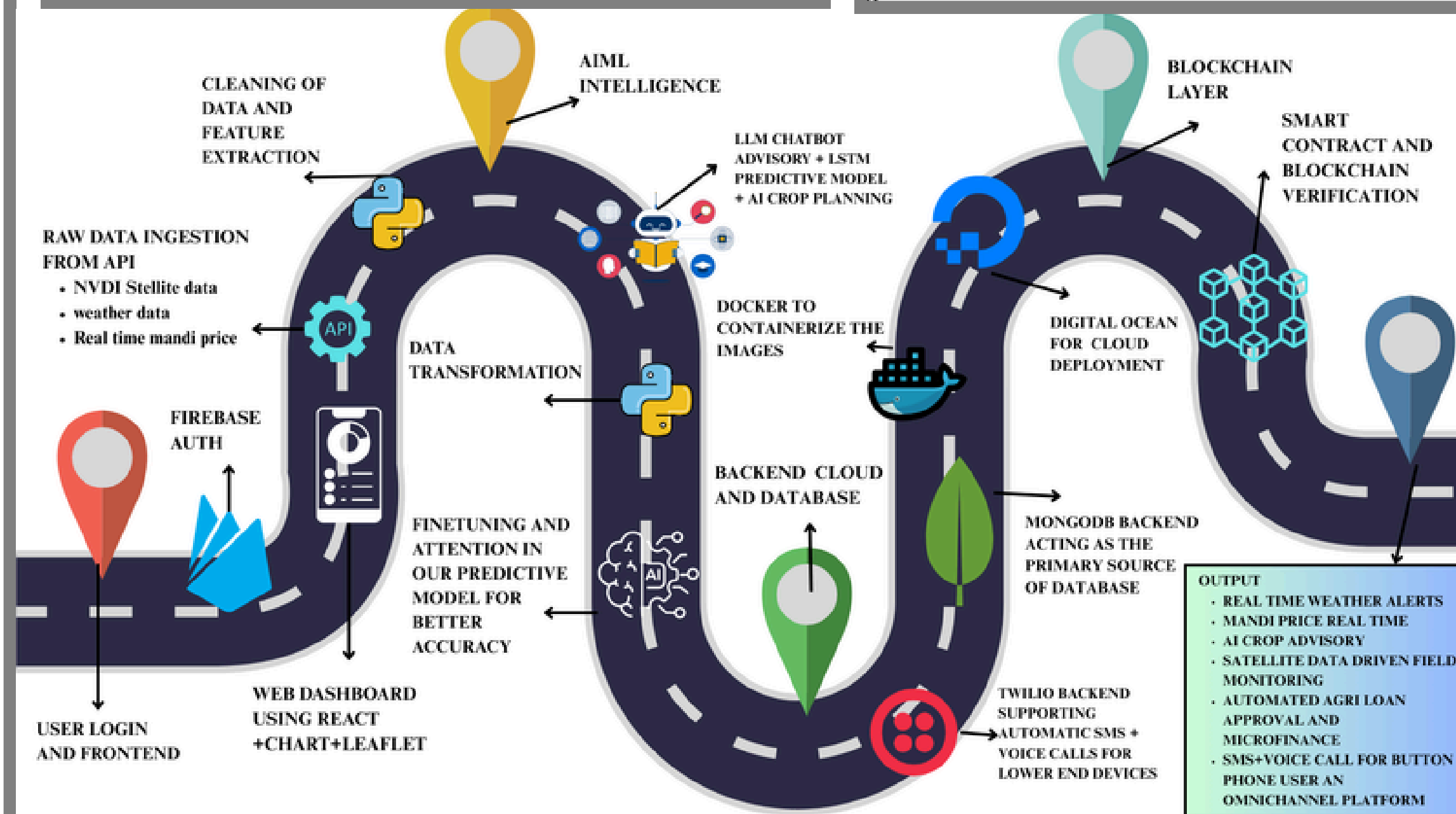
1.Smart Contracts and Blockchain verification

Algorithm Development:

- Data acquisition and preprocessing
- Integrate Data pipelines and APIs into an unified platform
- Machine Learning , Generative AI and Blockchain integration

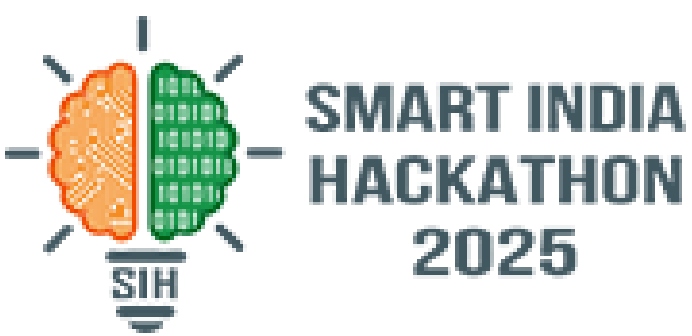
Tech Stack

React.js | Tailwind | Leaflet.js | Chart.js | Three.js | MongoDB | Docker | Firebase | Metamask | Node.js | Express.js | Python | APIs | Twilio | Figma | Framer | LLM | DigitalOcean | Hyperledger Blockchain |





FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA:

Technical Feasibility

- API integration Complexity
- ML Model Integration
- Low latency Processing

Financial Feasibility

- LSTM avoids costly APIs
- Twilio SMS ensures affordable digital inclusion
- .Smart contracts reduces transaction costs.

Market Feasibility

- Simple dashboards for insights.
- SMS access ensures inclusion in rural markets.
- Insurance automation addresses farmer trust issues.

Operational Feasibility

- Simplified omnichannel communication
- MongoDB supports efficient large-scale data handling.
- Blockchain nodes maintain secure decentralized operations.

POTENTIAL CHALLENGES AND RISKS

Technical

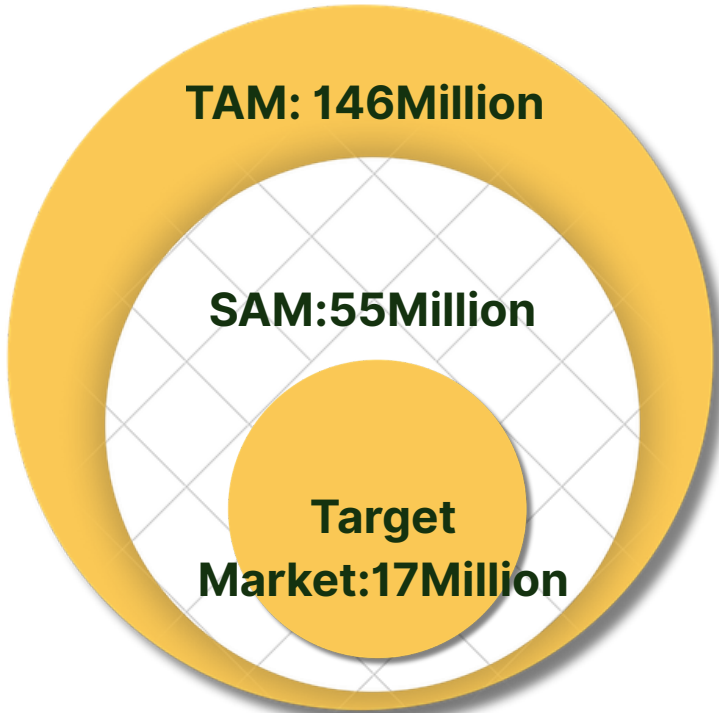
- ML Model Performance
- Scalability for large dataset

Financial

- Unforeseen Costs
- Revenue model sustainability

Market

- Market Saturation
- User Acceptance



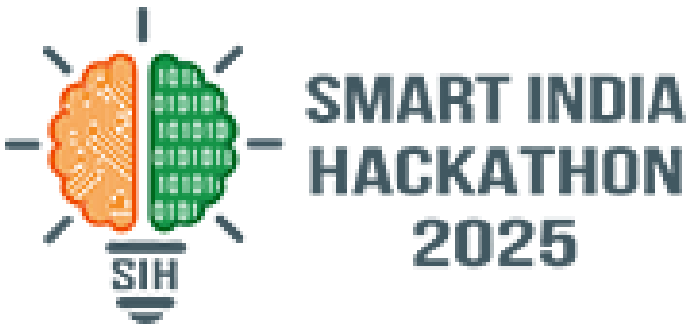
STRATEGIES TO OVERCOME THESE CHALLENGE

- Train models on diverse, high-quality data and use multiple reliable sources.
- Optimize the system for lower-end hardware and offer cloud-based processing options.





IMPACT AND BENEFITS

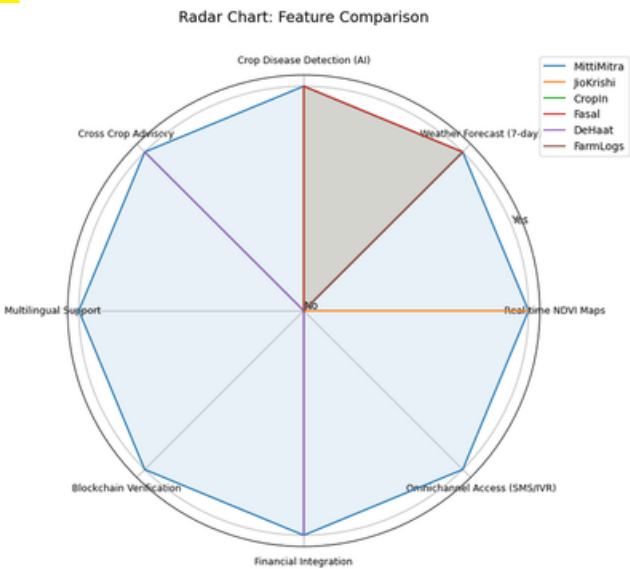


IMPACT ON TARGET AUDIENCE

BENEFITS

Enhancing the Agri-Market: Creates a transparent and efficient digital marketplace, reducing information gaps and ensuring fair pricing for produce.

Empowering Farmers: Shifts farmers from reactive guessing to proactive, data-driven decision-making for crop planning, resource management, and sales.



SOCIAL BENEFITS

- **Digital Inclusion:** Overcomes language, literacy, and technology barriers.
- **Community Empowerment:** Improves quality of life with easy access to vital services.

ECONOMIC BENEFITS

- **Higher Farmer Income:** Boosts profits through increased yields and fair pricing.
- **Reduced Risk:** Minimizes crop and financial losses with proactive alerts.
- **Lower Institutional Costs:** Cuts operational overhead for banks and insurers.

TECHNICAL BENEFITS

- **Secure & Scalable:** A robust platform fusing AI, Blockchain, and Satellite tech.
- **Guaranteed Trust:** Ensures tamper-proof data integrity with Blockchain.
- **Future-Ready:** Modular architecture allows for easy future expansion.

USE CASE

COMPARISION ANALYSIS

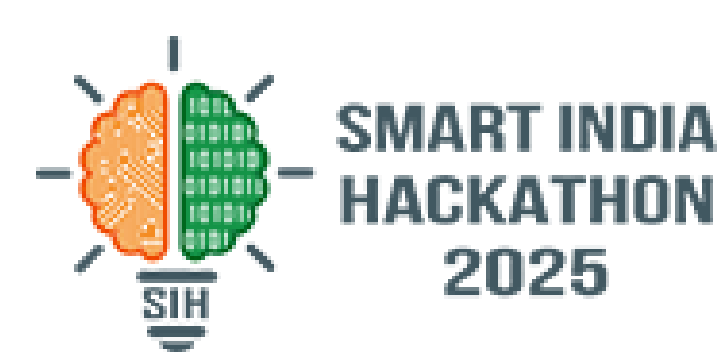
Feature / Platform	MittiMitra (Our Model)	JioKrishi	CropIn	Fasal	DeHaat	FarmLogs (Global)
Real-time NDVI Maps	✓	✓	✗	✗	✗	✗
Weather Forecast (7-day)	✓	✗	✓	✓	✓	✓
Crop Disease Detection (AI)	✓	✗	✓	✓	✗	✗
Cross Crop Advisory	✓	✗	✗	✗	✓	✗
Multilingual Support	✓	✗	✗	✗	✗	✗
Blockchain Verification	✓	✗	✗	✗	✗	✗
Financial Integration	✓	✗	✗	✗	✓	✗
Omnichannel Access (SMS/IVR)	✓	✗	✗	✗	✗	✗

Transforming Agri-Finance: Revolutionizes insurance and lending sectors with automated, transparent, and fraud-proof processes, making finance more accessible and viable.

Enabling Smart Governance: Provides government bodies with real-time, authenticated data for effective policy-making, resource allocation, and subsidy distribution.



RESEARCH AND REFERENCES



Research Papers Reference

1.Crop recommendation system for precision agriculture

Proceedings: ICoAC 2017, IEEE conference

DOI: 10.1109/ICoAC.2017.7951740 CoLab

Focus: Using data mining to analyse soil and other abiotic factors, helping farmers decide the appropriate crop based on soil requirements.

Relevance: Agriculture + AI advisory (crop choice).

2.Artificial Intelligence and Internet of Things for Sustainable Farming and Smart Agriculture

Publisher: IEEE Access (or close) – an IEEE paper. Bohrium

Focus: Framework combining IoT and AI for sustainable agricultural ecosystems. Covers sensor data, architecture, issues of scalability, etc.

Our Prototype Figma

<https://www.figma.com/proto/FI4MHnCwFRfnYuNIPEu5iN/MITTI-MITRA?node-id=0-1&t=XzHuYH2vaFjCjcQ1-1>

Explanation Video

https://youtu.be/D9tT44dm_y0?si=guCWABmQhsXFJjcR

Our Prototype Website

<https://infosysfin.vercel.app>

More detailed Explanation

https://www.canva.com/design/DAGy-ncb4l4/QFanrQtxvzl1eViCB71VcA/edit?utm_content=DAGy-ncb4l4&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

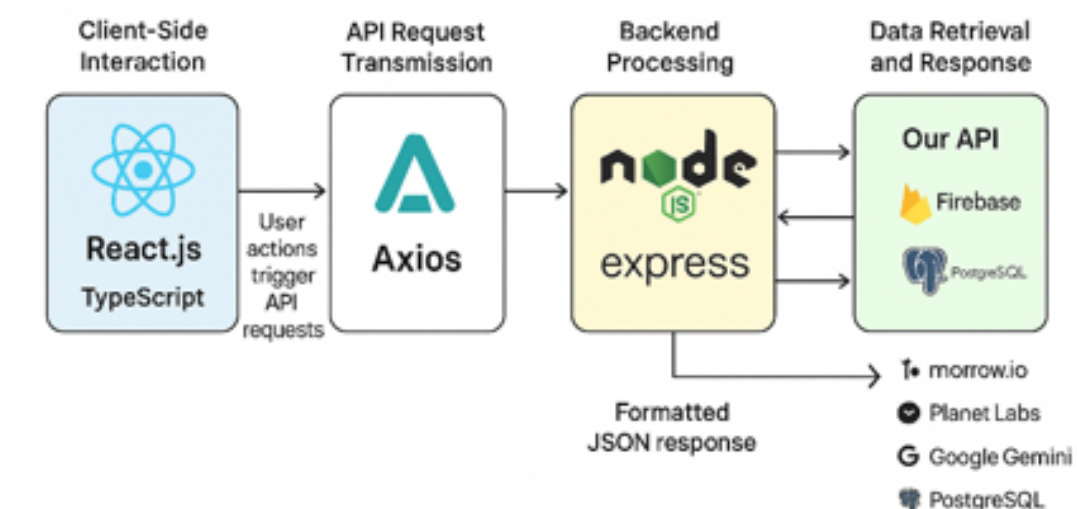
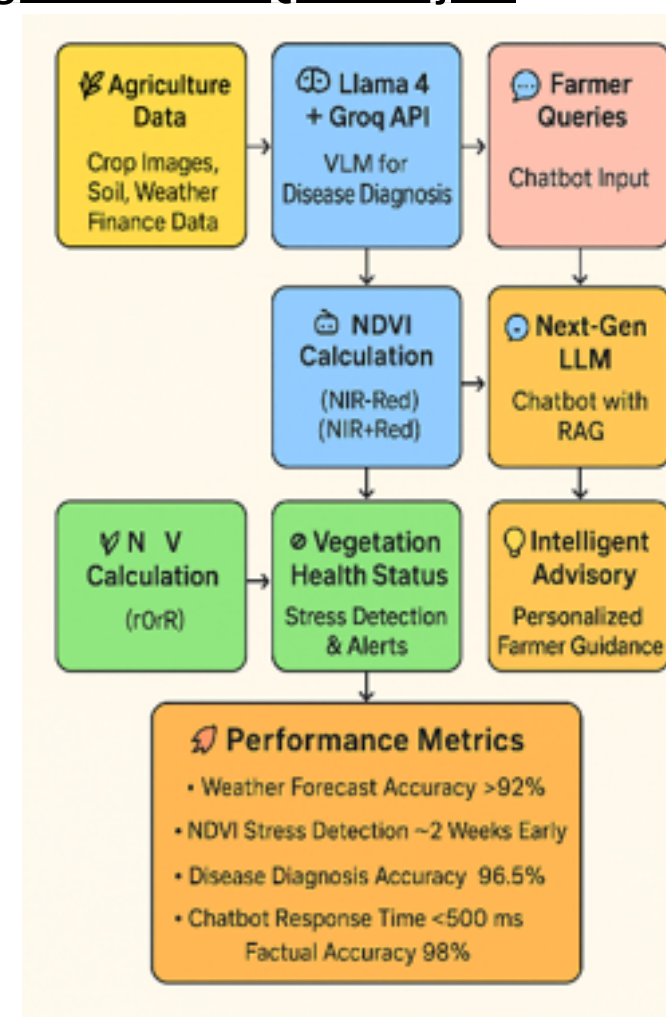


Table.2. Overall Analysis of model benchmark and F1 scores

Model		F1 Score	Accuracy / Benchmark Score
LLaMA 4	Maverick	94.5%	MMLU-Pro: ~80.5% - GPQA: ~69%
GPT-4o		84.4%	MMLU: 88.7%
DeepSeek-V3		91.6%	MMLU: 87.1% -

Workflow diagram of MittiMitra

Performance metrics of our VLM model and other metrics